

Ensemble Model Analytical Report on Organic Products

Business Understanding

In our analytical report on our client's customer data associated with organic products, we set out to answer the question, "Can the supermarket improve predictive accuracy by comparing different models and utilizing ensemble techniques to identify customers likely to purchase organic products?" Our dataset is composed of a target variable labeled "TargetBuy" which indicates customer loyalty to organic products, along with nine predictor variables that represent relevant attributes of the customers such as demographic features and purchasing behaviors.

To answer the business question, we will be using ensemble model analysis with the JMP software, which combines predictions from multiple models in order to develop a single prediction. Ensemble models are known for usually being more accurate than the individual models if the individual models are significantly different from each other and are in disagreement with each other, and therefore are worth generating and analyzing in order to potentially improve the supermarket's predictive accuracy. We will also be comparing our models using appropriate model fit statistics, such as accuracy, misclassification rate, profit-to-loss balance, and the Kolmogorov-Smirnov (KS) statistic to help select the model that has the overall best predictive accuracy.

Data Understanding

Figure 1 – Data Dictionary

Variable Name	Type	Sample Value	Description
ID	ID	140	Unique customer identifier
DemAffl	Input	10	Affluence score
DemAge	Input	76	Age of the customer
DemClusterGroup	Input	C	Demographic cluster group (A-F)
DemGender	Input	U	Gender (M=Male, F=Female, U=Unknown)
DemReg	Input	Midlands	Region where customer is located
DemTVReg	Input	Wales & West	TV region for targeted advertising
PromClass	Input	Gold	Promotion classification (Gold, Silver, Tin, etc.)
PromSpend	Input	16000	Amount spent on promotions
PromTime	Input	4	Time spent in promotions
TargetBuy	Target	0	Target binary variable indicating whether a purchase was made (1=Yes, 0=No)

In this figure, we see a data dictionary for our ID variable, nine predictor variables, and target variable, with data types and descriptions. We identify 22,213 data points in total.

Figure 2 – Key Statistics for Target and Predictor Variables



In Figure 2 above, we see the key statistics and visualizations for our nine predictor variables, and lastly, for our target variable. We note that PromSpend and PromTime have distributions that are skewed right, or skewed towards the higher extreme values. We note that the proportions of data in the classes for the categorical variables, such as DemGender and DemTVReg, are disproportionate, but not to an extent where the analysis would be affected.

Figure 3 – Data Quality Assessment

Variable Name	Number of Missing/ Unknown Values	Number of Outliers
DemAffl	0	0
DemAge	0	0
DemClusterGroup	728	0
DemGender	1746	0
DemReg	0	0
DemTVReg	0	0
PromClass	0	0
PromSpend	0	88
PromTime	0	1
TargetBuy	0	0

In this figure, we use the JMP software to identify missing or unknown values as well as outliers for our predictor variables and our target variables. We see that there were missing or unknown values for the variables DemClusterGroup and DemGender, as well as outliers for PromSpend and PromTime to be addressed in the data preparation step.

Data Preparation

We began our data preparation process by removing the few rows with missing or unknown values, identified in Figure 3, reducing our total number of data points to 19,788. This ensured that missing or unknown values would not have any influence on any of our models.

We then imputed the outliers with the average value of each respective continuous variable instead: the outliers in PromSpend were imputed with the average value of PromSpend, which was 4408.5043, and the 1 outlier in PromTime was imputed with the average value of PromTime, which was 6.5629169. This helped to normalize the distributions for both continuous variables, and helped prevent the models' predictions from being skewed by the outliers.

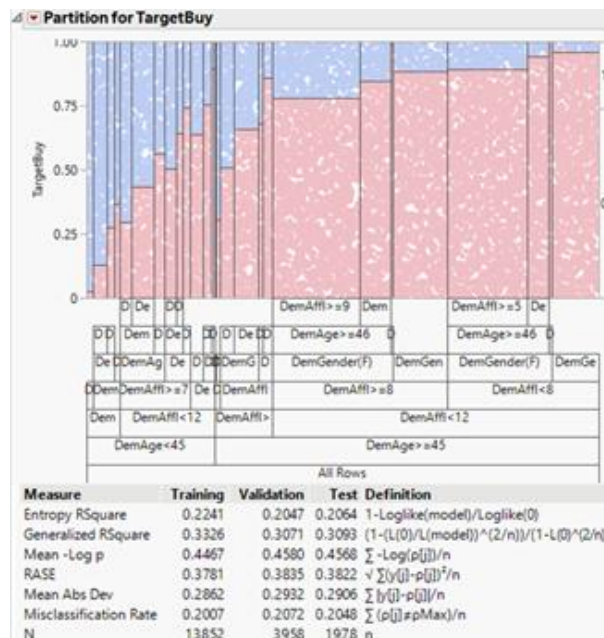
We also encoded the categorical variables, generating a dummy variable for each of our five categorical predictor variables. These dummy variables are named after the original variables, with a “_D” attached to the label, and give numerical values for the categorical values in the original variables. This encoding step was done so that we could include the categorical variables into our models that require all input variables to be of the continuous type.

Finally, we split our data into training, validation, and test sets, with 70% of the data in the training set, 20% of the data in the validation set, and 10% of the data in the training set. We used a stratified split that was based on our target variable to generate this split.

Modeling

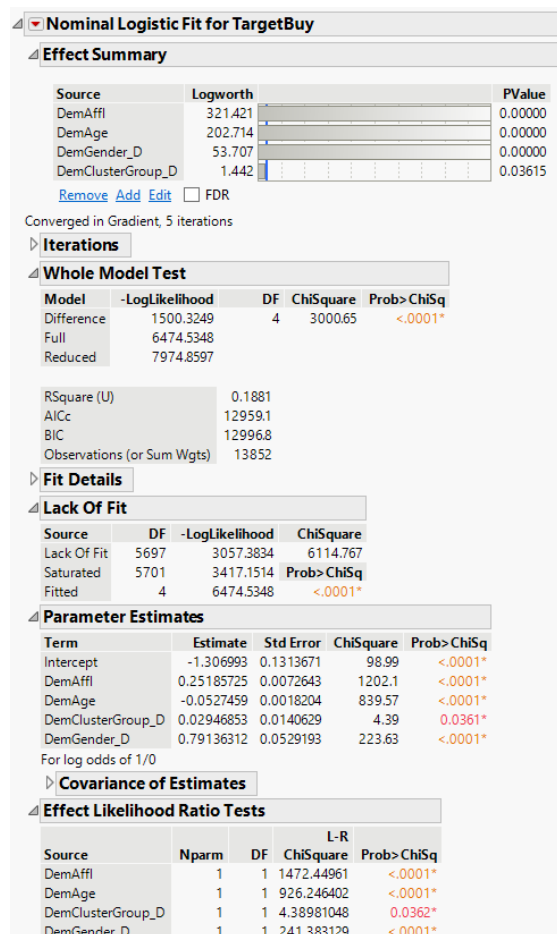
For our modeling step, we generate a decision tree model, a logistic regression model, and a neural network model, and an ensemble model through combining them afterwards.

Figure 4 – Decision Tree Model



In this figure, we see the decision tree model generated using the nine predictor variables and the target variable, and using the validation column to generate splits. This was a full variable decision tree, taking into account all of our predictor variables, in order to maximize the amount of information used to generate the predictions.

Figure 5 – Stepwise Logistic Regression Model



In this figure, we see the stepwise logistic regression model generated using the nine predictor variables and the target variable, and using the validation column to generate splits. We used stepwise regression to obtain the logistic regression model that best fits our data, with maximum efficiency, when taking model complexity into account. The stepwise regression model used a p-value threshold, with a probability to enter of 0.1 and a probability to leave of 0.05, in order to ensure a standard degree of statistical significance for each of the selected predictor variables. We selected the variables in a mixed direction to combine the forward and backward selection methods, and using the “Whole Effects” setting to include the effects associated with significant terms.

Figure 6 – Neural Network Model

Neural	
Validation Column: Validation	
Model Launch	
Model NtanH(3)	
Training	Validation
TargetBuy	TargetBuy
Measures	Value
Generalized RSquare	0.3223684
Entropy RSquare	0.2162751
RASE	0.3799493
Mean Abs Dev	0.2894305
Misclassification Rate	0.2023535
-LogLikelihood	6250.0962
Sum Freq	13852
Confusion Matrix	Confusion Matrix
Actual	Predicted Count
TargetBuy	0 1
0	9555 660
1	2143 1494
Confusion Rates	Confusion Rates
Actual	Predicted Rate
TargetBuy	0 1
0	0.935 0.065
1	0.589 0.411

In this figure, we see the neural network model generated using the nine predictor variables and the target variable, and using the validation column to generate splits. We used all of our predictor variables in order to maximize the information used to generate our predictions.

Figure 7 – Ensemble Model

Neural	
Validation Column: Validation	
Model Launch	
Model NtanH(3)	
Training	Validation
TargetBuy	TargetBuy
Measures	Value
Generalized RSquare	0.3355134
Entropy RSquare	0.2263474
RASE	0.3772922
Mean Abs Dev	0.2860168
Misclassification Rate	0.1990326
-LogLikelihood	6169.7708
Sum Freq	13852
Confusion Matrix	Confusion Matrix
Actual	Predicted Count
TargetBuy	0 1
0	9482 733
1	2024 1613
Confusion Rates	Confusion Rates
Actual	Predicted Rate
TargetBuy	0 1
0	0.928 0.072
1	0.557 0.443

In this figure, we see the ensemble model created using a neural network. This ensemble model was compiled using the prediction formulas from the three independent prediction models as predictors, in order to combine the predictive abilities of the three models. The response variable was still set to the target variable, and the validation column was used to split our data for training, validation, and testing.

Evaluation & Model Comparisons

The decision tree model had the advantage of maximizing input information by including all of the predictor variables, but had the disadvantage of increasing model complexity. The logistic regression model was more efficient due to selecting for only the most statistically significant predictors, but had a lower R^2 value, indicating that it fit the data more poorly. The neural network model also had the advantage of maximizing the input data by including all of the predictor variables, but at the cost of increased model complexity. The ensemble model had the advantage of combining the predictions from all three other models, but had the disadvantage of having a conditional accuracy, where the accuracy would only be improved if the individual models had sufficient disagreement.

Figure 8 – Model Comparison

Measures of Fit for TargetBuy											
Creator	.2	.4	.6	.8	Entropy RSquare	Generalized RSquare	Mean -Log p	RASE	Mean Abs Dev	Misclassification Rate	N
Partition					0.2185	0.3252	0.45	0.3796	0.2881	0.0000	19788
Fit Nominal Logistic					0.1857	0.2815	0.4688	0.3877	0.3026	0.2024	19788
Neural Model NTanH(3) - Save Formulas					0.2110	0.3155	0.4542	0.3815	0.2913	0.2034	19788
Neural Model NTanH(3) - Save Formulas					0.2208	0.3283	0.4486	0.3789	0.2879	0.2001	19788

In this figure, we see the results from running a model comparison process in JMP. The ensemble model, represented by the neural model in the bottom row, has the highest R^2 value at 0.3283, indicating that it accounts for the most of the variability in our data, with the lowest misclassification rate, testifying to the model's accuracy. This conclusion is consistent with our own analysis, where the ensemble model had the highest R^2 value out of the four models, with comparable misclassification rates of roughly 0.2 for all of the four models. Thus the ensemble model should be the best at predicting the customers that are likely to purchase organic products.

Scoring & Deployment

Figure 9 – Scoring New Data

ID	DemAffl	DemAge	DemClusterGroup	DemGender	DemReg	DemTVReg	PromClass	PromSpend	PromTime	Pred TargetBuy Ensemble
1	5	46	C	F	Midlands	Wales & West	Platinum	\$24,878.35	9	0
2	5	54	A	F	Midlands	Midlands	Tin	\$0.01	4	0
3	6	67	C	M	Midlands	Wales & West	Tin	\$0.01	8	0
4	5	38	C	F	South East	London	Silver	\$3,000.00	5	0
5	9	59	E	F	Scottish	N Scot	Gold	\$6,000.00	5	0
6	7	42	D	F	Midlands	Midlands	Tin	\$0.01	8	0
7	12	60	C	U	South East	S & S East	Tin	\$0.01	4	•
8	6	39	D	F	South East	London	Tin	\$0.01	5	0
9	9	42	E	F	South East	London	Silver	\$250.00	12	0
10	24	35	C	F	North	N West	Silver	\$1,000.00	5	1

In this figure, we score new customer data consisting of 10 data points. Because the customer with ID 7 had DemGender marked as “U” for “Unknown”, we were unable to generate a prediction for this customer. However, we did score the nine other customers in this dataset.

Deployment

We recommend based on our analysis, in response to the business question, that our client use model comparison based on appropriate model fit statistics and ensemble techniques to improve the predictive accuracy of identifying customers likely to purchase organic products. When we have several individual models that are significantly different from each other, an ensemble model is likely to have an improved accuracy compared to the individual models. Model comparison is also useful in improving predictive accuracy, because the utilization of appropriate model fit statistics, such as misclassification rate and R^2 value, can be used to identify the model that best fits the data. To direct advertising efforts towards customers likely to purchase organic products, ensemble techniques and model comparison should be used to help improve predictive model accuracy.